MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 10 (25-29 Oct 2026) - Period 1	Ch 8 - 8.6 Boolean operators + Task 8.2

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 10 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

CASE exit check: 85% correct; choosing CASE vs IF improving. Today: AND, OR, NOT and Programming task 8.2.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use logical operators AND, OR, NOT in conditions, evaluate them with truth tables, and complete Programming task 8.2. Syllabus 8.1.4(f); links to Topic 10 logic; AO2, AO3.

Success criteria — *what will successful learning look / sound like?*

I can evaluate compound conditions; I can write conditions with AND/OR/NOT in pseudocode and Python; my task 8.2 program handles all cases.

Key knowledge / skill / understanding:

AND needs both true; OR needs at least one; NOT reverses; brackets control order.

Likely misconception / access or challenge need:

'x > 5 AND < 10' is valid; OR means exactly one is true.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Gamification **Time:** 8 min

What students will do:

'Stand up if TRUE': teacher reads compound conditions about the class (wears glasses AND born in October...); students stand or sit, then write the truth table.

Evidence tool: Oral questioning

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Condition list; truth table slide.

Support (low performers / SEN):

Truth table with 2 rows filled.

Stretch (fast learners / gifted):

Write a condition that is TRUE for exactly 3 students.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Project-based task **Time:** 22 min

What students will do:

Programming task 8.2 (coursebook): plan, write pseudocode and Python using compound conditions, e.g. ticket pricing with age AND student status; test with a test table.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) Programming task 8.2; test table.

Support (low performers / SEN):

Pseudocode skeleton with conditions to complete.

Stretch (fast learners / gifted):

Add NOT and explain an equivalent condition without it.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 7 min

What students will do:

Pairs swap test tables and check each other's expected vs actual results; flag one missed test case.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Peer checklist.

Support (low performers / SEN):

Checklist with example.

Stretch (fast learners / gifted):

Write the missing test case.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: a theme-park ride - you can ride IF height >= 140 AND age >= 10. Model AND/OR/NOT truth tables, then pseudocode IF Age >= 10 AND Height >= 140 THEN ... and Python and/or/not. Link to logic gates (Topic 10).

What is new or innovative in this lesson? *(ETEC looks for this)*

Whole-body truth-table game makes logic visible; explicit link to logic gates topic builds cross-topic understanding.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Writing incomplete conditions (Age > 5 AND < 10); OR used instead of AND; missing brackets with NOT. Likely: 'OR means one or the other but not both'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: evaluate 3 compound conditions; fix 'Mark > 50 AND < 70'. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Finish task 8.2 and upload code + test table. Read coursebook 8.7 iteration (count-controlled).

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Standing game got everyone involved, including quieter students; 80% completed task 8.2. Compound condition errors dropped after the exit ticket.

What will I change next time?

Keep game to 6 minutes; extend testing time. Pre-teach the 'incomplete condition' error.

Students needing follow-up:

10BR2: 3 students need help with planning before coding; pair them with confident coders.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
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ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 10 (25-29 Oct 2026) - Period 2	Ch 8 - 8.7 Iteration: count-controlled loops

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 10 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Task 8.2: 80% submitted; test tables better. Today: FOR...TO...NEXT loops, STEP, and Python range().

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students write count-controlled loops (FOR ... TO ... STEP ... NEXT) in pseudocode and Python for/range(), and trace them. Syllabus 8.1.4(c); AO2.

Success criteria — *what will successful learning look / sound like?*

I can write a FOR loop with and without STEP; I can match it to Python range() correctly (end value!); I can trace a loop in a trace table.

Key knowledge / skill / understanding:

Count-controlled = fixed number of repeats; range(1, 11) gives 1-10; STEP changes the increment.

Likely misconception / access or challenge need:

range(1,10) includes 10; the loop variable can't be used inside; STEP must be 1.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Brainstorming **Time:** 6 min

What students will do:

List everyday tasks repeated a known number of times (5 daily prayers, 10 push-ups, 30 students' marks). Decide start, end and step for each.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Whiteboards.

Support (low performers / SEN):

Picture prompts.

Stretch (fast learners / gifted):

Find a task that is repeated an unknown number of times.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Code along: times-table printer, countdown with STEP -1, sum of 1..N; then write pseudocode FOR loops and convert to Python.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.7; syllabus pseudocode guide p.45; Python.

Support (low performers / SEN):

Templates with range() hints.

Stretch (fast learners / gifted):

Print a multiplication grid using a loop inside a loop (nested preview).

ACTIVITY 3 *student task + evidence of learning*

Strategy: Problem solving **Time:** 9 min

What students will do:

Trace table challenge: trace a FOR loop with STEP 2 and an accumulator; check with the Python debugger.

Evidence tool: Exit ticket

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Trace table sheet; Thonny debugger.

Support (low performers / SEN):

Partially completed trace table.

Stretch (fast learners / gifted):

Predict the output if STEP is -2.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: write 'I will not talk in class' 100 times - or use a loop. Model FOR Count <- 1 TO 10 ... NEXT Count and Python for count in range(1, 11). Use Thonny's step debugger to watch the counter change.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students watch loop variables change live in the debugger, connecting trace tables to real execution.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Off-by-one with range(); changing the loop variable inside; forgetting NEXT; indentation errors. Likely: 'my loop runs 9 times'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: write a FOR loop that outputs 5, 10, ... 50; give range() for 1 to 20. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.7 count-controlled questions; program: output the sum of the first N odd numbers.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Debugger made loops visible; trace tables improved. Students' everyday examples were rich and personal.

What will I change next time?

Explain range end value with a number line first; most errors were off-by-one.

Students needing follow-up:

10BR1: 2 students absent - send the loop worksheet. Gifted students ready for nested loops.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
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